$\frac{dn}{ds} = n(kb - t\tau)$ $\frac{db}{ds}$ τn

ALIAH UNIVERSITY

END SEMESTER EXAMINATION (AUTUMN) - 2023

COURSE CODE: MATUGBS01 COURSE NAME: ENGINEERING MATHEMATICS I

(FOR ECE, EEN, CSE, BCA, MEN, CEN)

Group A (40 Marks)

1. Answer the following Questions: 1X5=5
   (a) Define a sequence of real numbers.
   (b) Show that the series $1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \cdots$ is convergent and find its sum.
   (c) Define the radius of convergence of a power series.
   (d) Define a Beta function.
   (e) If $\Gamma(n)(n > 0)$ denotes the Gamma function, then show that $\Gamma(n+1) = n\Gamma(n)$.

2. Answer any one of the following questions: 5X1=5
   (a) If $x_n = \frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \cdots + \frac{1}{n(n+1)}$, then show that the sequence $\{x_n\}$ is bounded and monotonically increasing. Also find the limit of the sequence $\{x_n\}$.
   (b) If $x_n = \frac{4n+3}{3n+4}$, then show that the sequence $\{x_n\}$ is bounded above and monotonically increasing. Find limit of the sequence $\{x_n\}$.

3. Answer any one of the following questions: 5X1=5
   (a) Test the convergence of the following series:
   $$1 + \frac{1}{2}.\frac{1}{3} + \frac{1.3}{2.4}.\frac{1}{5} + \frac{1.3.5}{2.4.6}.\frac{1}{7} + \cdots$$
   (b) Use Cauchy's root test to examine the convergence of the series $\sum u_n$, where $u_n = \left\{\left(\frac{n+1}{n}\right)^{n+1} - \frac{n+1}{n}\right\}^{-n}$.

4. Answer any one of the following questions: 5X1=5
   (a) Evaluate $\lim_{x\to 0} \frac{x(5^x-1)}{1-\cos x}$.
   (b) If $f(x) = \begin{cases} x+1, & when \quad x \le 1, \\ 3 - ax^2, & when \quad x > 1, \end{cases}$
   then for what value of a , the function $f(x)$ will be continuous at $x = 1$.

5. Answer any one of the following questions: 5X1=5
   (a) if $y = e^{ax}\sin bx$, then show that $y_n = (a^2+b^2)^{\frac{n}{2}} e^{ax} \sin\left(bx + n\tan^{-1}\frac{b}{a}\right)$, where $y_n = \frac{d^n y}{dx^n}$.
   (b) If $y = \frac{1}{x^2+a^2}$, then show that $y_n = \frac{(-1)^n n!}{a^{n+2}} sin^{n+1}\theta \sin(n+1)\theta$, where $y_n = \frac{d^n y}{dx^n}$.

6. Answer any one of the following questions: 5X1=5
   (a) If $y = \tan^{-1} x$, then prove the following:

(i) $(1+x^2)y_2 + 2xy_1 = 0$,

(ii) $(1+x^2)y_{n+2} + 2(n+1)xy_{n+1} + n(n+1)y_n = 0$.

(b) If $2x = y^{\frac{1}{m}} + y^{-\frac{1}{m}}$, then show that $(x^2-1)y_{n+2} + (2n+1)xy_{n+1} + (n^2-m^2)y_n = 0$.

7. Answer any one of the following questions: 5X1=5

(a) State and prove Lagrange's Mean Value theorem.

(b) Using L' Hospital's rule evaluate $\lim_{x\to 0}(\frac{\sin x}{x})^{\frac{1}{x}}$.

8. Answer any one of the following questions: 5X1=5

(a) If $I_n = \int_0^{\frac{\pi}{2}} \sin^n x\, dx$, then show that $I_n = \frac{n-1}{n} I_{n-2}$, and hence find the value of $\int_0^{\frac{\pi}{2}} \cos^6 x\, dx$.

(b) Show that $\int_0^{\infty} e^{-x^4} x^2\, dx \times \int_0^{\infty} e^{-x^4}\, dx = \frac{\pi}{8\sqrt{2}}$.

Group B (40 Marks)

Answer any four questions. **(10 x 4 = 40)**

1.a) Transform the equation $\sqrt{r} = \sqrt{a}\sin(\theta/2)$ to its Cartesian form.

b) Find the equation of the three planes passing through the points (3,1,1) ,(1,-2,3) parallel to the co-ordinate axes. (5+5)

2.a) Evaluate $\int_C f(x,y)dx$ where $f(x,y) = x^2 + y^3$ and the curve C is the arc of the parabola $y = x^2$ in the xy-plane from (0,0) to (1,1).

b) Find the equation of the cone whose vertex is the origin and which passes through the curve of intersection of the plane $lx + my + nz = p$ and the surface $ax^2 + by^2 + cz^2 = 1$. (5+5)

3.a) Evaluate $\int_0^a \int_0^{\sqrt{(a^2-y^2)}} (x^2+y^2)\, dydx$ by changing to polar co-ordinates, with the help of Jacobian.

b) If the co-ordinates of A, B, C, D are $(-1,-2)$ (7,4) , (4,8) and $(-4,2)$ respectively, then show that $ABCD$ is a rectangle. (7+3)

4. Define limit and continuity of a vector function. If derivative of $F(t)$ exist at $t = t_0$, then prove that $F(t)$ is continuous at $t = t_0$. Is the converse true? Justify your answer. (4+ 3+3)

5.a) Evaluate $\int_0^a \int_0^x \int_0^y x^3 y^2 z\, dz dy\, dx$.

b) Evaluate $\int_0^{\pi/2} \int_0^{\pi} \sin(x+y)\, dydx$. (7+3)

6. a) Find the volume of the tetrahedron formed by the four planes $lx + my + nz = p$, $lx + my = 0$, $my + nz = 0$ and $nz + lx = 0$.

b) State Serret-Frenet's Formulae. (7+3)

7. State Green's theorem for a plane. Verify the theorem for $\oint_C [(xy+y^2)dx + x^2 dy]$ where C is the closed curve of the region bounded by $y = x$ and $y = x^2$. (3+7)

....